MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3540 POX PROCESS WATER TANK 3540-TNK-001
DESIGN CALCULATION OF FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of **POX PROCESS WATER TANK 3540-TNK-001**
The dimensions and reinforcement are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications , codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah- Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in standard excel sheet. Stability check against Overturning & Sliding are done in the excel sheet. Base Pressure check has been done in the excel sheet. For strength calculation Bending moment & Shear Force values are calculate in the conventional method & designed in the standard excel sheet.

4. DESIGN REQUIREMENTS

4.1 Reference Codes & Books

Reference is made in this standard as indicated in STRUCTURAL ENGINEERING DESIGN CRITERIA Doc No. : MD-216-1000-EG-ST-CRT-1001-A03 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

4.2 Materials Property

A. Density

4.2.A.(i) Density of Dry Soil : 1.9 T/m^3
4.2.A.(ii) Density of Concrete : 2.5 T/m^3
4.2.A.(iii) Density of Water : 1.0 T/m^3

B. Concrete Grade

4.2.B.(i) Foundation and Structural Members: 35 Mpa Strength of 28 days
4.2.B.(ii) Blinding Concrete below foundation bases: 15 Mpa Strength of 28 days

C. Reinforcement Grade

4.2.C.(i) Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
4.2.C.(ii) Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

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D. Grout Grade

4.2.D.(i) Grout shall be 50 mm cementitious of Strength 40 MPa.

4.3 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	MinimumCove (mm)
Exposed to Soil	Foundation	50
	Wall	50
	Slab on Grade	25

5. LOAD

5.1 Primary Loads

CCD & POX PROCESS WATER STORAGE TANK 3540-TNK-001 foundation design the primary load values are taken from **Drg. No. : MD-216-3540-EG-ST-GAD-1001**

1 DEAD LOAD - Nominal Thickness (D)
2 ROOF PIPE LIQUID LOAD (F_{roof})
3 SHELL PIPE LIQUID LOAD (F_{shell})
4 LIVE LOAD (L)
5 TANK LIQUID LOAD - Operating ($F_{operating}$)
6 TANK LIQUID LOAD - Test (F_{test})
7 WIND LOAD ON SHELL (W_{shell})
8 WIND LOAD ON ROOF (W_{roof})
9 SEISMIC LOAD (S)

5.2 Load Combinations for Stress Calculation

LOAD CASE	LOAD COMBINATION
101	$D + F_{roof} + F_{shell} + F_{test}$
102	$D + F_{roof} + F_{shell} + F_{test} + L$
103	$D + F_{roof} + F_{shell} + F_{operating}$
104	$D + F_{roof} + F_{shell} + F_{operating} + L$
105	$D + F_{roof} + F_{shell} + F_{operating} + 0.75*L$
106	$D + F_{roof} + F_{shell} + F_{operating} + W$
107	$D + F_{roof} + F_{shell} + F_{operating} + 0.7*E$
108	$D + F_{roof} + F_{shell} + F_{operating} + 0.75*W + 0.75*L$
109	$D + F_{roof} + F_{shell} + F_{operating} + 0.75*(0.7E) + 0.75*L$
110	$0.6*D + W$
111	$0.6*D + 0.7*E$

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5.3 Load Combinations for Strength Calculation

LOAD CASE	LOAD COMBINATION
201	$1.4*(D + F_{roof} + F_{shell} + F_{test})$
202	$1.4*(D + F_{roof} + F_{shell} + F_{test}) + 1.7*L$
203	$1.4*(D + F_{roof} + F_{shell} + F_{operating})$
204	$1.4*(D + F_{roof} + F_{shell} + F_{operating}) + 1.7*L$
205	$1.2*(D + F_{roof} + F_{shell} + F_{operating}) + 0.5*L$
206	$1.2*(D + F_{roof} + F_{shell} + F_{operating}) + 0.8*W$
207	$1.2*(D + F_{roof} + F_{shell} + F_{operating}) + 0.5*L + 1.6*W$
208	$1.2*(D + F_{roof} + F_{shell} + F_{operating}) + 0.5*L + E$
209	$0.9*D + 1.6*W$
210	$0.9*D + E$

6. SAFETY FACTOR AGAINST STABILITY

6.1 Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0
6.2 Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

7. ALLOWABLE SOIL BEARING CAPACITY

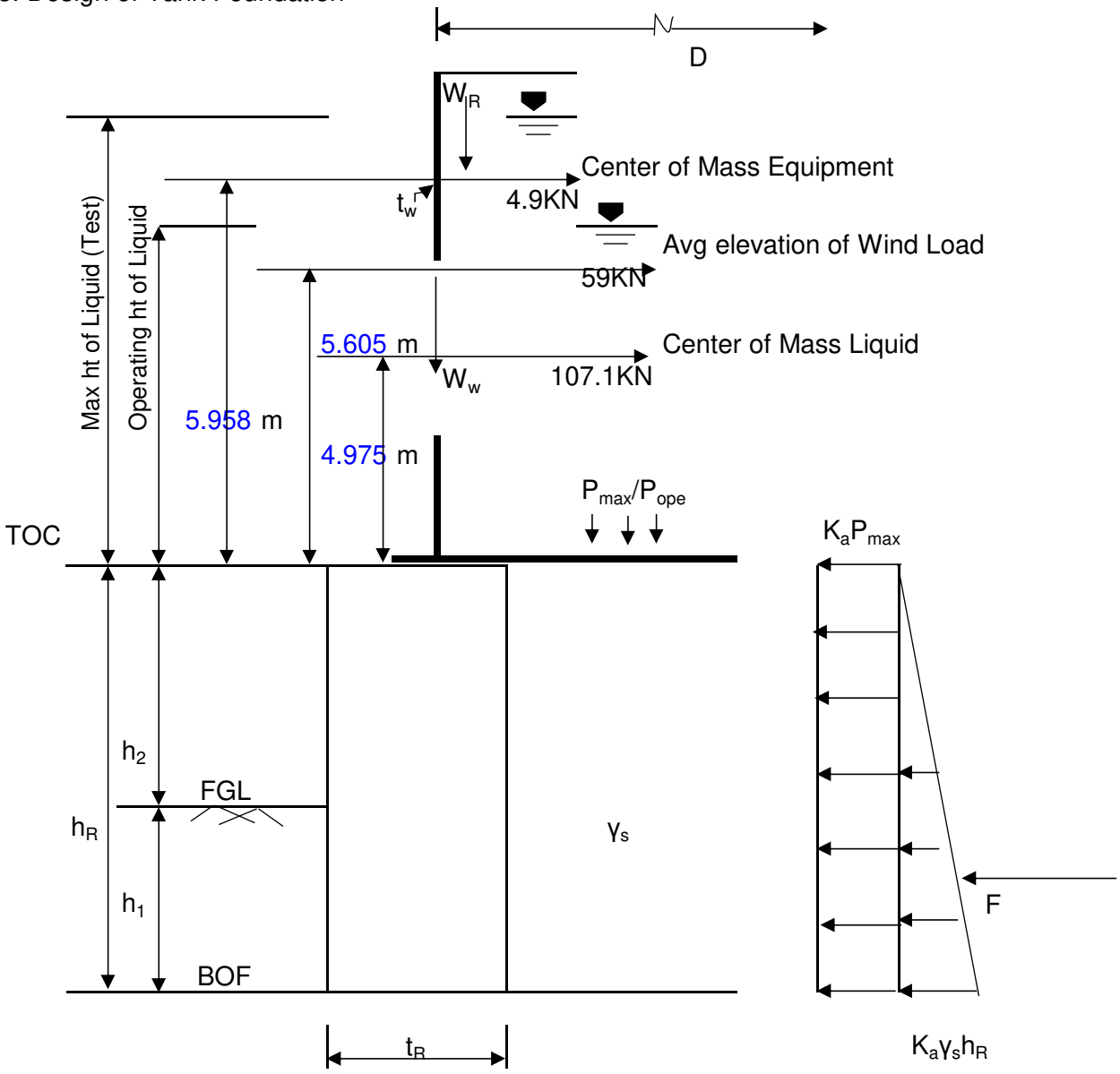
The net safe bearing capacity of soil for foundations design as obtained from page 39 of Doc. No. MD-216-0000-EM-GM-RPT-0003 is 670 KN/m2 at 1.2m depth below FGL
The neareast Bore Hole of POX PROCESS WATER TANK 3540-TNK-001 is BH-8, BH-13 & BH-14.
The net safe bearing capacity of soil as per approved L&T soil test report (Table-6 of Doc. No. MD-216-3000-EG-CV-RPT-1008-A02) is 425 KN/m2 for BH-8 at 1.2m depth below FGL & 900 KN/m2 for BH-13, BH-14 at 1.2m depth below FGL

However for safe design purpose we have considered the Net Safe Bearing Capacity at founding level as **375 KN/m2**.

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8. Design of Tank Foundation



Typical Section Through Tank

Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	10	m
Self weight of tank =	351	KN
Self weight of tank per m of circumference,W_w =	11.2	KN/m
External platform weight =	12	KN
Self weight of tank roof per m of circumference,W_R=	0.38	KN/m
Roof pipe liquid load (F_{roof}) =	53	KN
Shell pipe liquid load (F_{shell}) =	53	KN
Live Load (L) =	151	KN
Maximum weight of liquid (F_{test}) =	8475	KN
Pressure at base of tank , P_{max} = (F_{test}+F_{shell}+F_{roof})/A =	109	KN/m^2
Operating weight of liquid ($F_{operating}$) =	7672	KN
Pressure at base of tank , $P_{operating}$ = ($F_{operating}$+F_{shell}+F_{roof})/A =	99	KN/m^2

Foundation Material :

Concrete, f_c' =	35	MPa
Reinforcement, f_y =	420	MPa
Unit weight of concrete,γ_c =	24	KN/m^3

Soil Data :

Unit weight of soil,γ_s =	19	KN/m^3	(Natural sand)
Allowable bearing pressure, q_{all}=	375	KN/m^2	
Coefficient of active earth pressure,K_a =	0.33		

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	1.8	m
FGL to Bottom of foundation ,h_1 =	1.2	m
FGL to Top of concrete ,h_2 =	0.6	m
Ringwall thickness, t_R =	550	mm
Inner dia of ringwall,D_i =	9.5	m
Outer dia of ringwall,D_o =	10.6	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	41.49	m^3

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Check of soil bearing pressure (Test Liquid Load)

Wind force = 59 KN acting at a height of 7.405 m from BOF

M_{OT} = Overturning moment = 436.9 KN-m

M_{OT} / S_{xx} = 10.5 KN/m^2

Load Combination : 101. $D + F_{roof} + F_{shell} + F_{test}$

$q_{RF} = (W_w + W_R)/t_R + P_{max} + h_R\gamma_c$ = 173 KN/m^2 **< 375KN/m2, O.K.**

Load Combination : 102. $D + F_{roof} + F_{shell} + F_{test} + L$

$q_{RF} = (W_w + W_R)/t_R + P_{max} + h_R\gamma_c + L/A$ = 175 KN/m^2 **< 375KN/m2, O.K.**

Load Combination : 106. $D + F_{roof} + F_{shell} + F_{test} \pm W$

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 184 KN/m^2 **< 375KN/m2, O.K.**

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 163 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : 110. $0.6*D \pm W$

$q_{RE} = (W_w + W_R)/t_R + h_R\gamma_c$ = 64.2 KN/m^2

$q_{max} = 0.6*q_{RE} + M_{OT}/S_{xx}$ = 49.1 KN/m^2 **< 375KN/m2, O.K.**

$q_{min} = 0.6*q_{RE} - M_{OT}/S_{xx}$ = 27.99 KN/m^2 **No (-)ve pressure**

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 436.9 KN-m

M_{OR} = Restoring moment = (351+PI()*10*0.55*1.8*25)*10/2 = 5642.7 KN-m

FOS = M_{OR} / M_{OT} = 5642.72 / 436.9 = 12.92 **> 1.5, Hence O.K.**

Check for soil bearing pressure (Operating Liquid Load):

Seismic force at center of mass equipment = 351/(351+7672)*112 = 4.9 KN
acting at a height of 7.758 m from BOF

Seismic force at center of mass slurry = 7672/(351+7672)*112 = 107.1 KN
acting at a height of 6.775 m from BOF

M_{OT} = Overturning moment = 764 KN-m

M_{OT} / S_{xx} = 18.4 KN/m^2

Load Combination : 103. $D + F_{roof} + F_{shell} + F_{operating}$

$q_{RF} = (W_w + W_R)/t_R + P_{operating} + h_R\gamma_c$ = 163 KN/m^2 **< 375KN/m2, O.K.**

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Load Combination : 104. $D + F_{roof} + F_{shell} + F_{operating} + L$

$q_{RF} = (W_w + W_R)/t_R + P_{operating} + h_R\gamma_c + L/A$ = 165 KN/m^2 **< 375KN/m2, O.K.**

Load Combination : 107. $D + F_{roof} + F_{shell} + F_{operating} \pm 0.7*E$

$q_{max} = q_{RF} + 0.7*M_{OT}/S_{xx}$ = 176 KN/m^2 **< 375KN/m2, O.K.**

$q_{min} = q_{RF} - 0.7*M_{OT}/S_{xx}$ = 150 KN/m^2

Load Combination : 111. $0.6*D \pm 0.7*E$

$q_{max} = 0.6*(W_w + W_R)/t_R + 0.6*h_R\gamma_c + 0.7*M_{OT}/S_{xx}$ = 51.4 KN/m^2

$q_{max} = 0.6*(W_w + W_R)/t_R + 0.6*h_R\gamma_c - 0.7*M_{OT}/S_{xx}$ = 25.6 KN/m^2 **No (-)ve pressure**

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 75.1 KN/m

Hoop tension in ringwall, $T = 1/2FD$ = 375 KN

Factored design tension, T_u= 525 KN FOS = 1.4

$A_{st,reqd} = T_u/\phi f_y$ = 1390 mm^2 ϕ = 0.9

$A_{st,min} = 0.002* h_R t_R$ = 1980 mm^2

$A_{st,reqd}$ (per face of ring wall)= 990 mm^2

Required : 12 dia bar @ 206 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{st,\ provided}$(per face) = 1358 mm^2 **O.K.**

Vertical Reinforcement :

$A_{st,min} = 0.0012* t_R$ = 660 mm^2/m

$A_{st,reqd}$ (per face of ring wall)= 330 mm^2

Required : 10 dia bar @ 238 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{st,\ provided}$(per face) = 393 mm^2 **O.K.**

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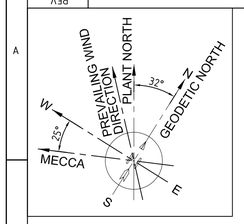
ATTACHMENT - 1 : Mechanical Input Drawing.

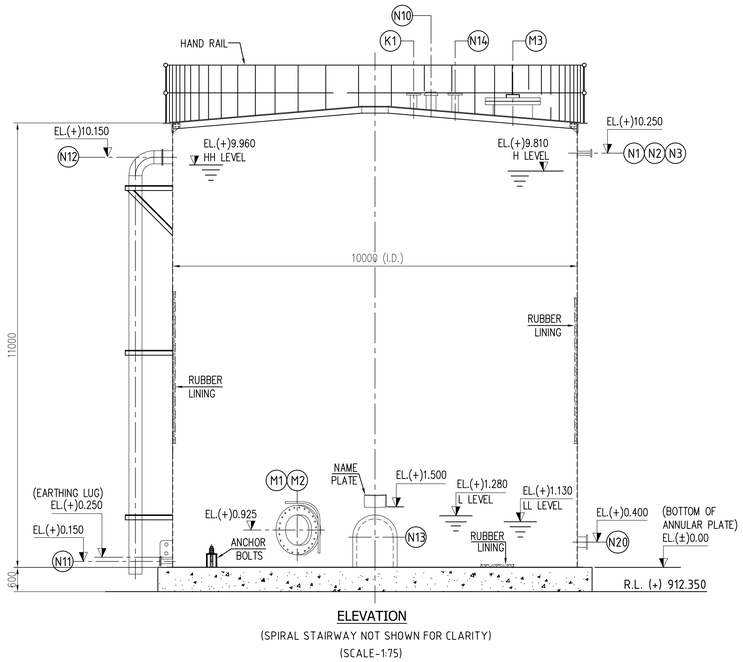
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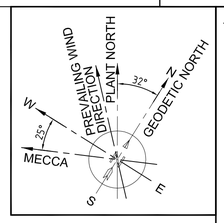


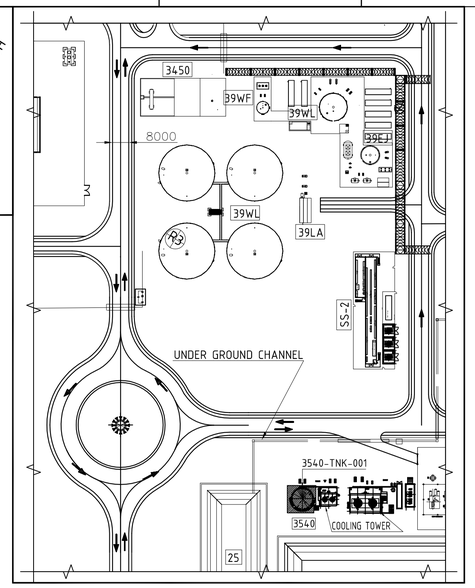


ELEVATION

(SPIRAL STAIRWAY NOT SHOWN FOR CLARITY)

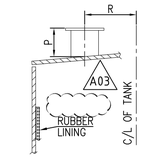
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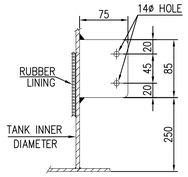


KEY PLAN

SCALE 1:1500

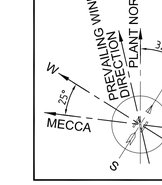


ROOF NOZZLES / FITTINGS

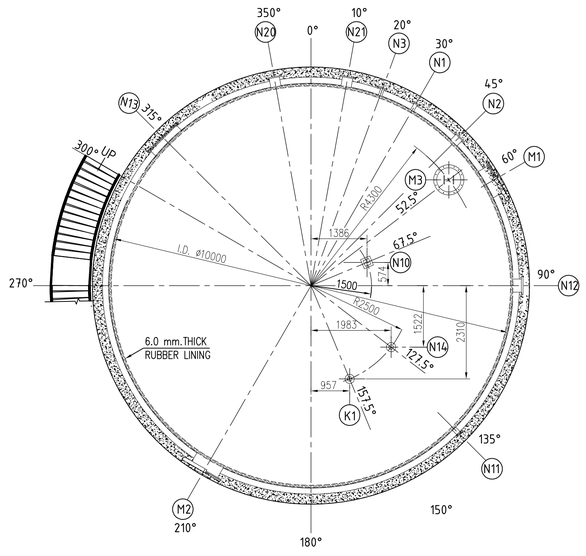


DETAIL OF EARTHING LUGS

(QUANTITY : 2 Nos.)



SHELL NOZZLES / FITTINGS



PLAN

3540-TNK-001

(SCALE-1:75)

DESIGN DATA

LOCATION	MANSOURAH - MASSARAH GOLD PROJECT
EQUIPMENT TAG NO.	3540-TNK-001
TANK NAME	POX PROCESS WATER TANK
FIELD ERECTED/SHOP FABRICATED	FIELD ERECTED
DESIGN CODE & EDITION	API 650, 12TH EDITION, ADDENDUM 3, AUGUST 2018

DESIGN DATA - PROCESS

STORED FLUID		POX PROCESS WATER
DENSITY OF CONTENT (t/m³)	NORMAL OPERATION / DESIGN	1.0 / 1.0
PRESSURE (kPa)	VACUUM / DESIGN	ATM / ATM
TEMPERATURE (°C)	NORMAL OPERATION / DESIGN	36 / 50
DENSITY OF SCALE (t/m³)		N/A
DENSITY SCALE THICKNESS (mm)		N/A
OPERATING LEVELS (mm)	MINIMUM / MAXIMUM	1120 / 9950
DESIGN FILL RATE (m³/h)		220 (TBC)
DESIGN EMPTY RATE (m³/h)		200 (TBC)
PWHT REQUIRED (YES/NO). IF YES SPECIFY REASON FOR PWHT		NO

DESIGN DATA - MECHANICAL

TANK DIAMETER (I.D.) (mm)		10000
TANK CYLINDRICAL HEIGHT (mm)		11000
NET WORKING CAPACITY (m³)		668
MAXIMUM CAPACITY (m³)		780
TANK HAS SLOPED BOTTOM	(YES/NO)	YES
	BOTTOM SLOPE	1/120 CONICAL APEX-UP
ROOF REQUIRED	ROOF TYPE	SUPPORTED CONE ROOF
	ROOF SLOPE	1 IN 12
	ROOF SUPPORT	SUPPORTED ON RAFTER/TRUSS
	ROOF UNIFORM LIVE LOAD(kPa)	1.2
AGITATOR / RAKE REQUIRED	(YES/NO)	NO
	EQUIPMENT TYPE(AGITATOR/RAKE)	N/A
	AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)	N/A
CORROSION ALLOWANCE (mm)	SHELL / ROOF / BOTTOM	1.6 / 0 /1.6
	STRUCTURAL	1.6

MATERIAL SPECIFICATION

SHELL, BOTTOM, ANNULAR, ROOF	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
COD	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
MANWAY NECKS/ MANWAY FLANGES	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A53 Gr.B/ ASTM A106 Gr.B
NOZZLE NECKS (PIPES) (> 600 NB)	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	FLAT RING 1.5mm/3.00mm THICK TANGED GRAPHITE
GASKETS FOR MANWAY/ COD	3mm THICK COMPRESSED NON-ASBESTOS SHEETS.
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	ASTM A36 GR.36 OR EN 10025 GR.S275 JR/S355 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 Gr. B
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
EARTHING LUGS	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
ANCHOR BOLTS	ASTM F1554
CURB ANGLE	ASTM A36 GR.36 OR EN 10025 GR.S275 JR/S355 JR
NAME PLATE	STAINLESS STEEL (SS 316)

DESIGN DATA - OTHERS

SEISMIC USE GROUP, SUG PER API650 E.3.1		SUG I
FOUNDATION TYPE		RING WALL
INSULATION REQUIRED	(YES/NO)	NO
	MATERIAL	N/A
	THICKNESS(mm) ROOF/SHELL/BOTTOM	N/A
LINER REQUIRED	(YES/NO)	YES (A03)
	LINER MATERIAL	NATURAL RUBBER SHORE HARDNESS 65±5A
	LINER THICKNESS (mm) ROOF/SHELL/BOTTOM	6.0 mm.
PAINTING	EXTERNAL	YES
	INTERNAL	YES (UNLINED AREA)

TANK APPURTENANCE (AS REQUIRED)

PIPE SUPPORT	YES
SPIRAL STAIRWAY	YES
ROOF CIRCUMFERENTIAL HANDRAIL	YES
WIND GIRDER	AS REQUIRED
CATHODIC PROTECTION	NO
LEAK DETECTION	NO
INTERNAL BAFFLES	NO
EARTHING LUGS	YES

NOZZLE TABLE

NOZZLE	SERVICE	SIZE (DN)	THICKNESS (SCHEDULE)	RATING #	TYPE	ELEVATION "H"	PROJECTION "P"	RADIUS "R"	ORIENTATION	LOCATION	REMARK
N1	INLET-DEMIN WATER	80	Sch XS	CL 150	RFSO	10.250	200	-	30°	SHELL	
N2	INLET-POX COOLING WATER	200	Sch XS	CL 150	RFSO	10.250	200	-	45°	SHELL	
N3	RE-CIRCULATION LINE	80	Sch XS	CL 150	RFSO	10.250	200	-	20°	SHELL	
N10	TANK VENT	150x150 (TBC)	-	RECTANGULAR WITH COVER ARRANGEMENT		-	300	1500	67.5°	ROOF	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300	-	135°	SHELL	
N12	OVERFLOW	300	Sch XS	CL 150	RFSO	10.150	250	-	90°	SHELL	
N13	CLEAN OUT DOOR (COD)-hxb	900x1200	-	AP1 650		-	-	-	315°	SHELL	
N14	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	400	2500	127.5°	ROOF	
N20	OUTLET-POX PROCESS WATER	250	Sch XS	CL 150	RFSO	0.400	250	-	350°	SHELL	
N21	OUTLET-POX PROCESS WATER	250	Sch XS	CL 150	RFSO	0.400	250	-	10°	SHELL	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	400	2500	157.5°	ROOF	
M1	MANHOLE C/W DAVIT	750	-	AP1 650		0.925	175	-	60°	SHELL	
M2	MANHOLE C/W DAVIT	750	-			0.925	175	-	210°	SHELL	
M3	MANHOLE C/W HINGED COVER	600	-			-	500	4300	52.5°	ROOF	

ABBREVIATIONS / LEGENDS

ID	- INNER DIAMETER
COD	- CLEAN OUT DOOR
EL	- ELEVATION
RF	- REINFORCEMENT
TBC	- TO BE CONFIRMED
U.O.S	- UNLESS OTHERWISE STATED
H	- HIGH
HH	- HIGH HIGH
L	- LOW
LL	- LOW LOW
RFSO	- RAISED FACE SLIP ON
C/W	- COUNTER WITH
FGL	- FINISH GROUND LEVEL
PWHT	- POST WELD HEAT TREATMENT

NOTE:-

1. ALL DIMENSIONS ARE IN MILLIMETER AND ELEVATION ARE IN METERS U.O.S.
2. ALL ELEVATION MEASUREMENTS ARE MEASURED FROM BOTTOM SIDE OF BASE PLATE.
3. SHELL NOZZLE ELEVATIONS ARE FROM BOTTOM OF BASE PLATE TO CENTERLINE OF NOZZLE.
4. SHELL NOZZLE PROJECTIONS ARE MEASURED FROM OUTSIDE OF SHELL TO FLANGE FACE.
5. ROOF NOZZLE PROJECTIONS ARE MEASURED FROM TOP OF ROOF SURFACE TO FLANGE FACE.
6. ROOF NOZZLE RADIUS MEASUREMENT ARE DISTANCES FROM TANK CENTERLINE TO NOZZLE CENTERLINE.
7. ALL NOZZLE LOCATION, ELEVATION AND ORIENTATION ARE INDICATIVE ONLY AND SHALL BE FINALIZED DURING DETAIL DESIGN PHASE OF PIPELINE ENGINEERING.
8. NOZZLE LOAD SHALL BE CONSIDERED & INCORPORATED AFTER COMPLETION OF PIPING ENGINEERING.
9. LOCATION OF EARTHING LUGS SHALL BE SHOWN IN SEPARATE DRAWING.
10. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY.
11. ANCHOR BOLTS SHOWN HERE IS INDICATIVE, QUANTITY & LOCATION SHALL BE SHOWN IN SEPARATE DRAWING.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A03	31/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS MARKED (A03)	DDS	SPU	SPU	KBJ	
MD-216-3540-EG-ME-SPC-1001	SPECIFICATION : POX PROCESS WATER TANK	A02	05/03/2020	REVISED AS PER EMPLOYER COMMENTS ON 12.02.2020	DDS	SPU	SPU	KBJ	
MD-216-3540-EG-PR-PID-1001	PIPING INSTRUMENTATION DIAGRAM (PID)			AND REVISED AS MARKED (A02)					
MD-216-3540-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM (PFD)	A01	15/01/2020	ISSUED FOR APPROVAL	SUG	SPU	SPU	KBJ	

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OUTOTEC DRAWING NUMBER: 900707__GAU01__AG02__MTL01__35401 | DOCUMENT ID: OU100667072 | REVISION: 2

DRAWN	SUG	02/01/2020
CHECKED	SPU	10/01/2020
ENGINEER	SPU	10/01/2020
APPROVED	KBJ	15/01/2020
EMPLOYER		

SCALE: 1:75 | A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CCD & POX PROCESS WATER STORAGE
POX PROCESS WATER TANK 3540-TNK-001

DRAWING NUMBER: MD-216-3540-EG-ME-GAD-1003 | REV: A03

ATTACHMENT - 1 : Structural Input Drawing.

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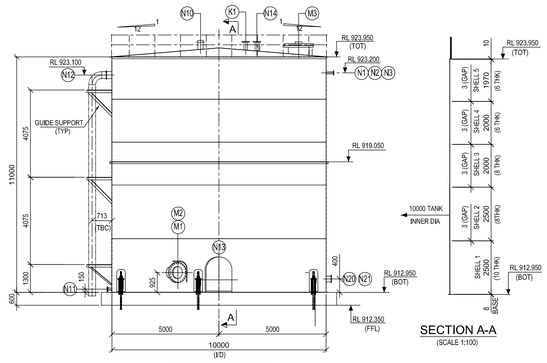
DRAWING NUMBER: MD-216-3540-EG-ST-GAD-1001 REV 001

MATERIAL SPECIFICATION

SHELL, BOTTOM, ANNULAR, ROOF	EN 10025 GR.S275 J0
COD	EN 10025 GR.S275 J0
MANWAY NECKS/ MANWAY FLANGES	EN 10025 GR.S275 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A53 Gr.B/ ASTM A106 Gr.B
NOZZLE NECKS (PIPES) (> 600 NB)	ASTM A36 GR.36 OR EN 10025 GR.S275 JO/S355 J0
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	FLAT RING 1.5 MM/3.00 MM THICK TANGED GRAPHITE
GASKETS FOR MANWAY/ COD	3mm THICK COMPRESSED NON-ASBESTOS SHEETS.
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	EN 10025 GR.S275 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	EN 10025 GR.S275 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	EN 10025 GR.S275 J0
EARTHING LUGS	EN 10025 GR.S275 J0
ANCHOR BOLTS	ASTM F1554 GR.36
CURB ANGLE	EN 10025 GR.S275 JR
NAME PLATE	STAINLESS STEEL (SS 316)

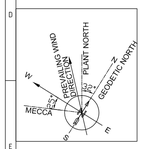
FOUNDATION LOAD DATA:

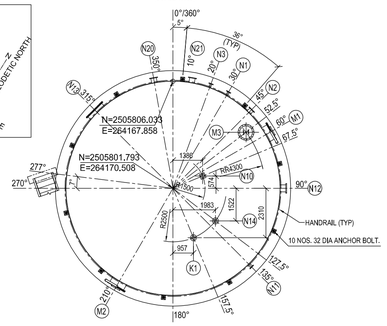
BASIC LOAD CASE	VERTICAL COMPRESSION (KN)	VERTICAL UPLIFT (KN)	HORIZONTAL SHEAR (KN)	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT) (KN-M)	SLAB MOMENT (KN-M)
DEAD LOAD (NOMINAL THICKNESS)	421	–	–	–	–
DEAD LOAD (CORRODED THICKNESS)	351	–	–	–	–
ROOF PIPE LIQUID LOAD	53	–	–	–	–
SHELL PIPE LIQUID LOAD	53	–	–	–	–
LIVE LOAD	151	–	–	–	–
TANK LIQUID LOAD (OPERATING)	7672	–	–	–	–
TANK LIQUID LOAD (TEST)	8475	–	–	–	–
WIND LOAD ON SHELL	–	–	59	336	–
WIND LOAD ON ROOF	–	78	–	392	–
SEISMIC LOAD	266	266	112	468	629



TANK ELEVATION
(SCALE 1:100)

SECTION A-A
(SCALE 1:100)





PLAN
(SCALE 1:100)

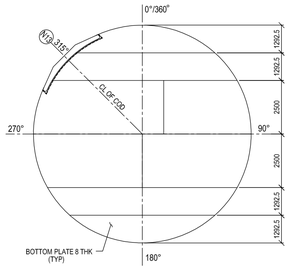
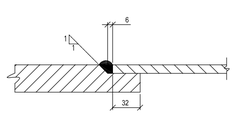
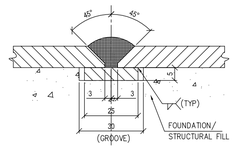


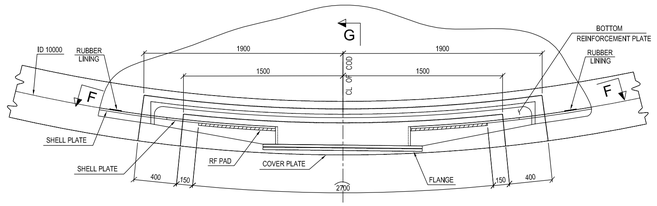
PLATE ARRANGEMENT OF BOTTOM PLATE
(SCALE 1:100)



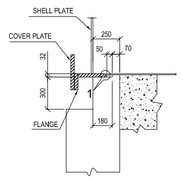
DETAIL-1
SCALE- 1:10



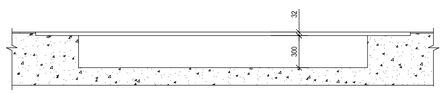
TYPICAL BOTTOM PLATE JOINT DETAIL
SCALE- 1:10



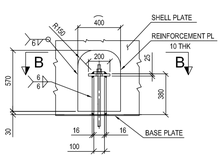
DETAIL OF CLEAN OUT DOOR (COD)
(SIZE 900x1200)
SCALE- 1:20



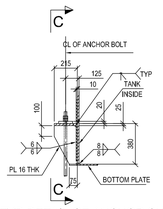
SECTION G-G
SCALE- 1:20



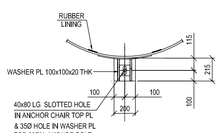
SECTION F-F
SCALE- 1:20



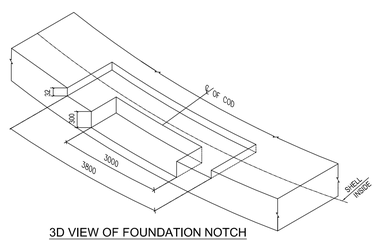
SECTION C-C
SCALE- 1:20



TYP DETAIL OF ANCHOR CHAIR
SCALE- 1:20



SECTION B-B
SCALE- 1:20

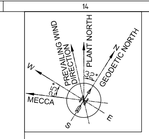


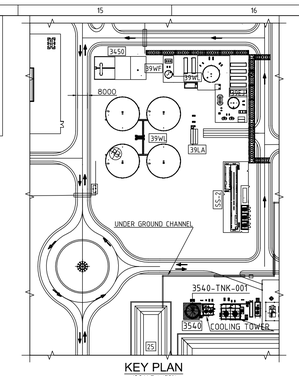
3D VIEW OF FOUNDATION NOTCH (FOR COD)

NOZZLE TABLE

NOZ.	SERVICE	SIZE (DN)	SCH	RATING #	TYPE	ELEV."H"	PROJ."P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA ★	RF PAD THK. ★	REMARK
N1	INLET-DEMIN WATER	80	Sch XS	CL 150	RFSO	10.250	200	-	30°	SHELL	265	6	
N2	INLET-POX COOLING WATER	200	Sch XS	CL 150	RFSO	10.250	200	-	45°	SHELL	485	6	
N3	RE-CIRCULATION LINE	80	Sch XS	CL 150	RFSO	10.250	200	-	20°	SHELL	265	6	
N10	TANK VENT	150x150 (TBC)	-	RECTANGULAR WITH COVER ARRANGEMENT	-	-	300	1500	67.5°	ROOF		6	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300	-	135°	SHELL	305	10	
N12	OVERFLOW	300	Sch XS	CL 150	RFSO	10.150	250	-	90°	SHELL	685	6	
N13	CLEAN OUT DOOR (COD)-hxb	900x1200	-	API 650		-	-	-	315°	SHELL	1372X2700	10	
N14	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	400	2500	127.5°	ROOF	305	6	
N20	OUTLET-POX PROCESS WATER	250	Sch XS	CL 150	RFSO	0.400	250	-	350°	SHELL	585	10	
N21	OUTLET-POX PROCESS WATER	250	Sch XS	CL 150	RFSO	0.400	250	-	10°	SHELL	585	10	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	400	2500	157.5°	ROOF	305	6	
M1	MANHOLE C/W DAVIT	750	-	API 650		0.925	175	-	60°	SHELL	1545	10	
M2	MANHOLE C/W DAVIT	750	-			0.925	175	-	210°	SHELL	1545	10	
M3	MANHOLE C/W HINGED COVER	600	-			-	500	4300	52.5°	ROOF	1255	6	

★ REFER NOTES NO. 12





KEY PLAN
(SCALE 1:1750)

DESIGN DATA

TANK DETAILS	LOCATION			MANSOURAH - MASSARAH GOLD PROJECT
	TANK NAME			POX PROCESS WATER TANK
	EQUIPMENT TAG No.			3540-TNK-001
	TANK DIMENSIONS			INSIDE DIAMETER : 10000mm, HEIGHT: 11000mm
	QUANTITY (NUMBER)			1
	ROOF TYPE & SLOPE			SLOPED CONE ROOF (SLOPE 1 IN 12) CONICAL APEX UP
	BOTTOM TYPE & SLOPE			SLOPED BOTTOM (SLOPE 1 IN 120) CONICAL APEX UP
	FIELD ERECTED/SHOP FABRICATED			FIELD ERECTED
	STORED FLUID			POX PROCESS WATER
DESIGN CODE	DESIGN CODE			API 650-TWELFTH EDITION, MARCH 2013, ADDENDUM 3, AUGUST 2018
OPERATING DATA	OPERATING TEMPERATURE			36° C
	NET WORKING CAPACITY			668 m^3 (TBC)
	MAXIMUM CAPACITY			780 m^3 (TBC)
	OPERATING LEVEL	MINIMUM		1120 mm (TBC)
		MAXIMUM		9950 mm (TBC)
	OPERATING DENSITY OF CONTENT			1.0 t/m^3
MECHANICAL & PROCESS DATA	DESIGN TEMPERATURE			50° C
	DESIGN PRESSURE (INTERNAL/EXTERNAL)			ATM/ATM
	TEST PRESSURE			ATM
	HYDRO STATIC TEST PRESSURE			FULL OF WATER
	DESIGN DENSITY OF CONTENT			1.0 t/m^3
	TESTING			AS PER API 650
	INSPECTION			AS PER API 650
	DENSITY OF SCALE			N/A
	DESIGN SCALE THICKNESS	TANK SHELL/ROOF		N/A
	DESIGN FILL RATE			220 m^3/hr (TBC)
	DESIGN EMPTY RATE			200 m^3/hr (TBC)
	CORROSION ALLOWANCE		SHELL	1.6 mm
			ROOF PLATE	0 mm
			BOTTOM PLATE / ANNULAR PLATE	1.6 mm
			ANCHOR BOLT	0 mm
			STRUCTURAL	1.6 mm
WIND LOAD DATA	BASIC WIND SPEED			43 m/sec. (155 Km/hr.)
	SITE EXPOSURE			C
	VELOCITY PRESSURE EXPOSURE COEFFICIENT			K_z = AS PER ELEVATION
	TOPOGRAPHIC FACTOR			K_{zt}=1.0
	WIND DIRECTIONALITY FACTOR			K_d=0.95
SEISMIC LOAD DATA	SITE CLASS			B
	SEISMIC USER GROUP			SUG - I
	IMPORTANCE FACTOR	(I)		1
	SCALING FACTOR	(Q)		2/3
	MAXIMUM DESIGN LIQUID LEVEL	(H)		0.960 m
	CO-EFFICIENT TO ADJUST THE SPECTRAL ACCELERATION FROM 5% TO 0.5% DAMPING	(K)		1.50
	SEISMIC CO-EFFICIENTS			S_s= 0.1g, S_1= 0.02g, F_a= 1.0, F_v= 1.0 S_{ds} = 0.067g. S_{d1}= 0.013g, T_s= 4 sec.
OTHER DATA	ROOF LIFE LOAD			1.2 kPa
	FOUNDATION TYPE (RING WALL/SLAB)			RING WALL
	PWHT REQUIRED(YES/NO)/ IF YES,SPECIFY REASON FOR PWHT			NO
	REASON FOR PWHT(PROCESS REASON/PER API CODE)			AS PER API 650
	RADIOGRAPHY			AS PER CL 8.1 - API 650
	AGITATOR RAKE REQUIRED	(YES/NO)		NO
		EQUIPMENT TYPE (AGITATOR/RAKE)		N/A
		AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)		N/A
	INSULATION		SHELL	N/A
			ROOF	N/A
	LINER REQUIRED	(YES/NO)		YES
		LINER MATERIAL		NATURAL RUBBER SHORE HARDNESS 65±5A
		LINER THICKNESS (ROOF/SHELL/BOT)		6.0 mm.
	PAINTING		INTERNAL	YES (UNLINED AREA) (REFER NOTE 17)
			EXTERNAL	YES
	PIPE SUPPORT			YES
	SPIRAL STAIRWAY			YES
	ROOF CIRCUMFERENTIAL HANDRAIL			YES
	WIND GIRDER			AS REQUIRED
	CATHODIC PROTECTION			NO
	LEAK DETECTION (APPURTENANCE)			NO
	INTERNAL BAFFLES			NO
	EARTHING LUGS			YES

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. BOLT HOLES IN NOZZLE FLANGES & MANHOLES SHALL STRADDLE VERTICAL CENTER LINES ON SHELL & NORTH SOUTH CENTERLINES ON ROOF.
5. NOZZLE PROJECTION ON SHELL SIDE REFERS FROM EXTERNAL SURFACE OF SHELL PLATE FACE OF FLANGE. NOZZLE PROJECTION ON ROOF SIDE REFERS FROM TOP OF ROOF PLATE TO FACE OF FLANGE.
6. VERTICAL SHELL JOINTS SHOULD PREFERABLY BE OFFSET BY 1000 MM.
7. CURB ANGLE JOINTS SHALL STAGGER ATLEAST 300 MM WITH VERTICAL SHELL PLATE WELD.
8. REINFORCEMENT PADS TO HAVE TELLTALE HOLES OF 6 MM AND SHALL BE TESTED AT 100 kPa PRESSURE WITH SOAP SOLUTION AND THEN FILLED WITH HARD GREASE.
9. FLANGE (COD & MANHOLE) GASKET SEATING SURFACE SHALL BE PROVIDED WITH 0.0032 TO 0.0063 mm (3.2 TO 6.3 µm) ROUGHNESS FINISH.
10. EARTHING LUG DETAIL & LOCATION TO BE CONFIRMED IN SEPARATE DRAWING.
11. SUITABLE NOTCHES AS PER API 650 TO BE PROVIDED IN THE FOUNDATION/GROUT FOR THE COD.
12. MINIMUM THICKNESS OF RF PAD, COLLAR, FLANGE AND COVER PLATES FOR NOZZLES ARE PROVIDED BASED ON HYDROSTATIC LOAD AND FINAL SIZE SHALL BE CONFIRMED AFTER NOZZLE DESIGN IN SEPERATE DRAWING.
13. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD-216-3540-EG-ST-GAD-1002 & 1003
14. INSTALLATION OF LINER ON TANK SHALL BE DONE AFTER COMPLETION OF SHELL WELDING. SEPARATE DOCUMENT WILL BE RELEASED FOR INSTALLATION OF LINER.
15. ALL FOUNDATION LOADS ARE "UNFACTORED".
16. TANK CO-ORDINATES MENTIONED IN THIS DRAWING ARE BASED ON AS DATE E3D LAYOUT. HOWEVER, FINAL CO-ORDINATES SHALL BE CROSS CHECKED BEFORE CASTING FOUNDATION.
17. PAINTING SHALL CONFORM TO MAADEN SPECIFICATION NUMBER MD-101-SMEM-EG-ME-SPC-0007 AND SEPARATE PAINTING DETAILS & APPLICATION PROCEDURE WILL BE SUBMITTED FOR APPROVAL.

LEGEND:

ID - INNER DIAMETER
OD - OUTER DIAMETER
COD - CLEAN OUT DOOR
LG - LONG
FGL - FINISHED GROUND LEVEL.
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
RL - REDUCED LEVEL
RF - REINFORCEMENT
TOC - TOP OF CONCRETE
GST - GUSSET
PL - PLATE
STIFF - STIFENER
THK - THICKNESS
CLT - CLEAT
UNO - UNLESS NOTED OTHERWISE
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
NOZ - NOZZLE
DN - DIAMETER NOMINAL
SCH - SCHEDULE
ELEV - ELEVATION
PROJ - PROJECTION
MM - MILLIMETER
TYP - TYPICAL
GA - GENERAL ARRANGEMENT
DRG - DRAWING
CL - CENTER LINE
TOS - TOP OF STEEL
WP - WORKING POINT
N/S - NEAR SIDE
F/S - FAR SIDE
DIA - DIAMETER
C/F - CUT FORM
TBC - TO BE CONFIRMED
TOG - TOP OF GRATING
N/A - NOT APPLICABLE
ATM - ATMOSPHERIC
MIN - MINIMUM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	22.11.2023	AS BUILT	SJH	PRI	GG	AG	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	C01	09/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER'S	SJH	PRI	GG	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA			COMMENTS ON 20/03/2020 & REVISED AS MKD.					
MD-216-3540-EG-ME-GAD-1003	POX PROCESS WATER TANK 3540-TNK-001	A01	26/02/2020	ISSUED FOR REFERENCE	AKDS	PRI	GG	AG	

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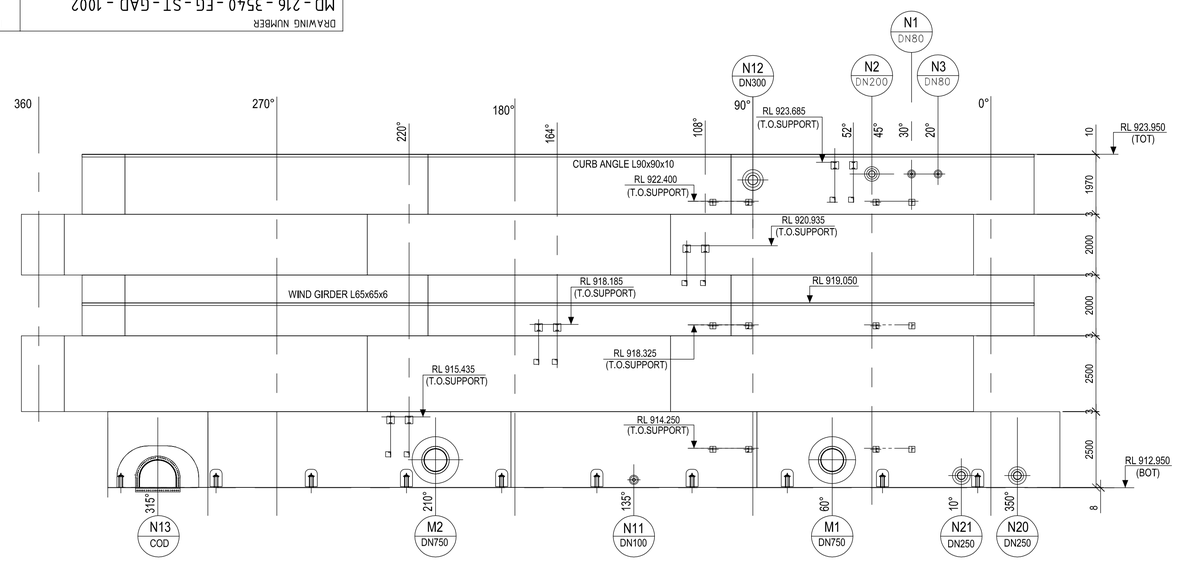
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LARSEN & TOUBRO DRAWING NUMBER: O19045_080_S_G_26140_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG02__MTL01__35402 — DOCUMENT ID: OU100673488 — REVISION: 2

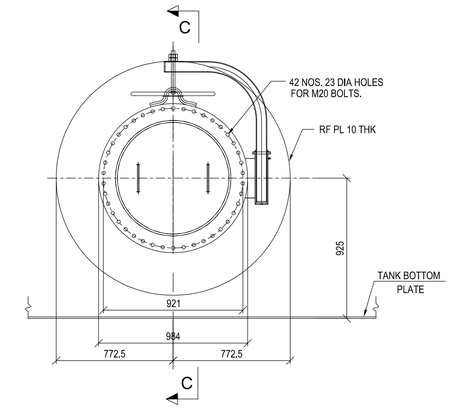
DRAWN	AKDS	21/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	PRI	25/02/2020	GENERAL ARRANGEMENT
ENGINEER	GG	24/02/2020	CCD & POX - PROCESS WATER STORAGE
APPROVED	AG	26/02/2020	PLAN, ELVN & DETAIL OF TANK 3540-TNK-001
EMPLOYER			
SCALE	1:100,20	A0	DRAWING NUMBER: MD-216-3540-EG-ST-GAD-1001 — REV: D01

DRAWING NUMBER MD-216-3540-EG-ST-GAD-1002 REV D01



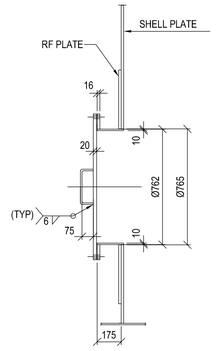
ELEVATION OF DEVELOPED SHELL PLATE (FOR TANK NO 3540-TNK-001)

(SCALE 1:100)



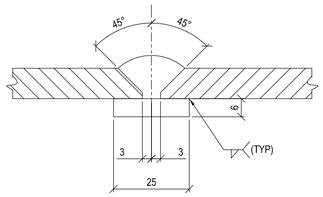
TYP. DETAIL OF SHELL DAVIT MANHOLE

SCALE- 1:20



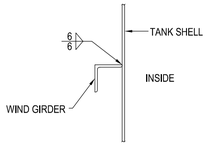
SECTION C-C

SCALE- 1:20



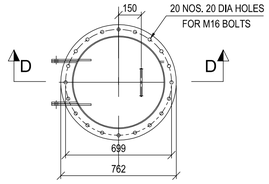
TYP.BOTTOM PLATE JOINT DETAIL

SCALE- 1:10

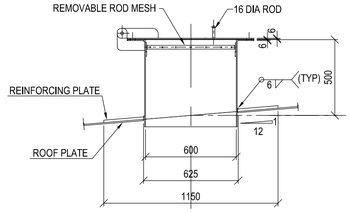


SECTION F-F

SCALE- 1:10

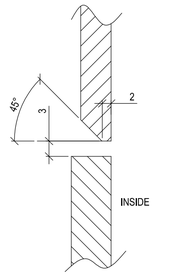


TYP. DETAIL OF ROOF MANHOLE (600 DN)



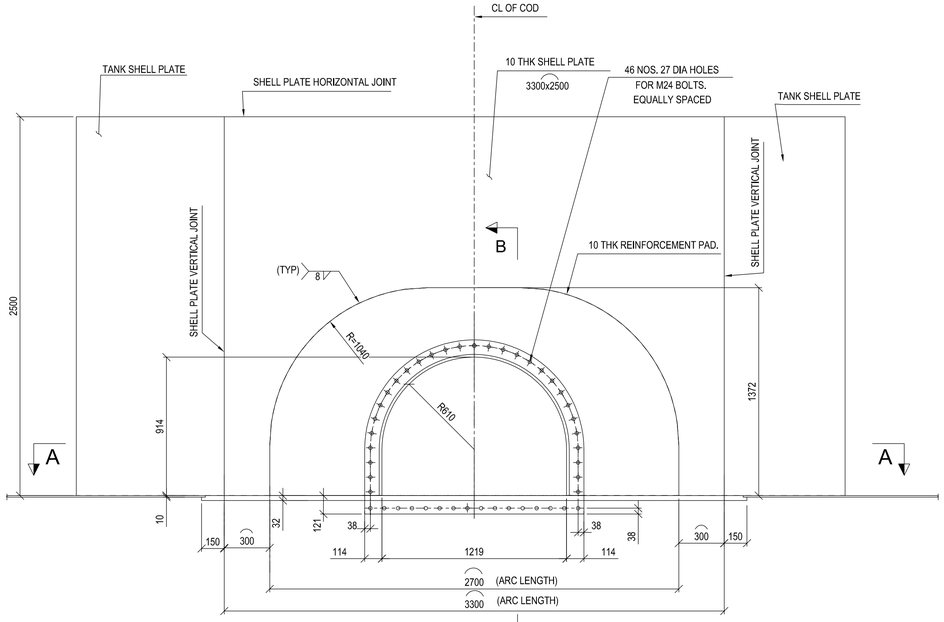
SECTION D-D

SCALE- 1:20



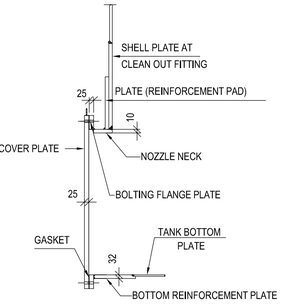
TYP.SHELL PLATE JOINT DETAIL

SCALE- 1:10



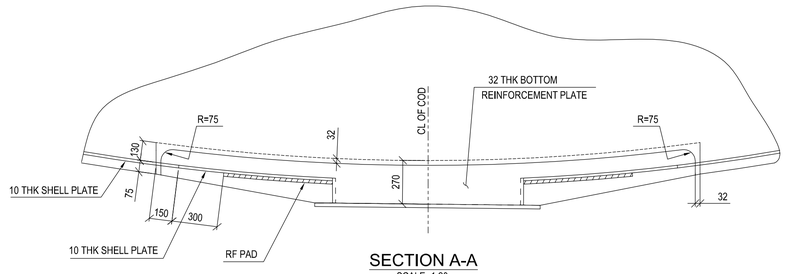
DETAIL OF CLEAN OUT DOOR (COD)

(SIZE 900x1200)
SCALE- 1:20



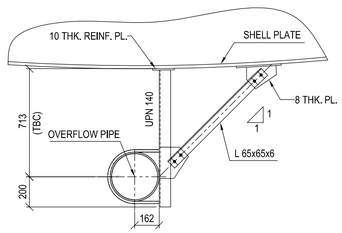
SECTION B-B

SCALE- 1:20



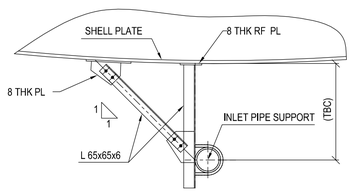
SECTION A-A

SCALE- 1:20



PLAN FOR OVERFLOW SUPPORT. FOR N12

SCALE- 1:20



PLAN FOR INLET POX COOLING WATER SUPPORT FOR N2

SCALE- 1:20

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
3. FOR ALL OTHER NOTES REFER DRAWING MD - 216 - 3540 -EG -ST - GAD - 1001.
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3540 -EG -ST - GAD -1001 & 1003.
6. FOR LEGEND LIST REFER DRAWING MD - 216 - 3540 -EG -ST - GAD - 1001.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	22.11.2023	AS BUILT	SJH	PRI	GG	AG	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	C01	09/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER'S	SJH	PRI	GG	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA			COMMENTS ON 20/03/2020 & REVISED AS MKD.					
MD-216-3540-EG-ME-GAD-1003	POX PROCESS WATER TANK 3540-TNK-001	A01	26/02/2020	ISSUED FOR REFERENCE	AKDS	PRI	GG	AG	

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OUTOTEC DRAWING NUMBER: 900707 _GAU01 _AG02_MTL01 _35403 — DOCUMENT ID: OU100673489 — REVISION: 2

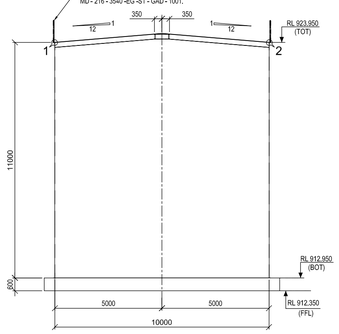
DRAWN	AKDS	21/02/2020
CHECKED	PRI	25/02/2020
ENGINEER	GG	24/02/2020
APPROVED	AG	26/02/2020
EMPLOYER		
SCALE	1:100, 20	A1

MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CCD & POX - PROCESS WATER STORAGE
SHELL PLT DETAIL OF TANK 3540-TNK-001

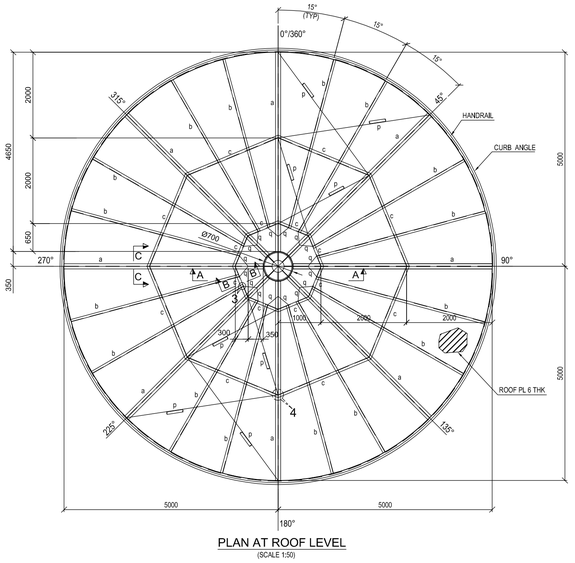
DRAWING NUMBER: MD - 216 - 3540 - EG - ST - GAD - 1002 — REV: D01

DRAWING NUMBER: MD-216-3540-EG-ST-GAD-1003 REV D01

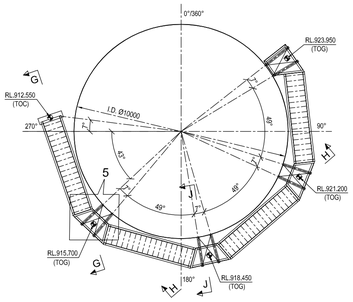
FOR LOCATION OF HANDRAIL REFER DRAWING NO. MD-216-3540-EG-ST-GAD-1001.



TANK ELEVATION
(SCALE 1:100)



PLAN AT ROOF LEVEL
(SCALE 1:50)

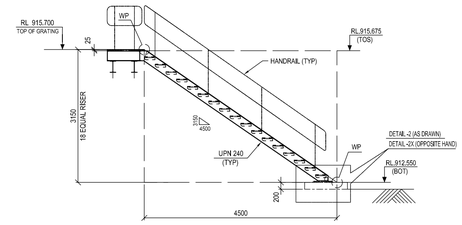


STAIR CASE PLAN
(SCALE 1:100)

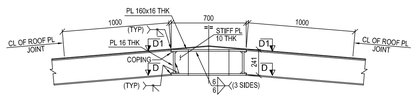
SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTION	FACTORED AXIAL FORCE(kN)	FACTORED SHEAR FORCE(kN)	REMARKS
a	IPE240	I	60	20	MATERIAL GRADE: S275 JR
b	UPN120	[	40	-	
c	UPN 140	[	70	-	
p	L 65X65X6	⌐	25	-	
q	L 50X50X6	⌐	45	-	
t	IPE200	I	-	-	
u	L 65X65X6	⌐	-	-	
v	L 65X65X6	⌐	-	-	
w	UPN240	[	-	-	

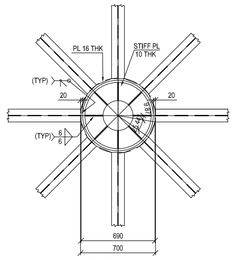
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED. MINIMUM 6 MM FILLET WELD OF 150 MM LENGTH SHALL BE PROVIDED. ALL GUSSET PLATE ARE 8 MM THK (UNO).



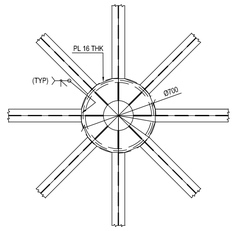
SECTION G-G
ELEVATION OF STAIR
(SCALE 1:50)



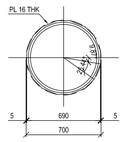
VIEW A-A
(TYP. CONNECTION OF MAIN RAFTER WITH CENTRAL DRUM)
(SCALE 1:20)



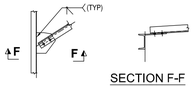
SECTION D-D
(SCALE 1:20)



SECTION D1-D1
(SCALE 1:20)

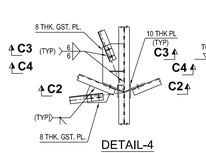


TOP PLATE
(SCALE 1:20)

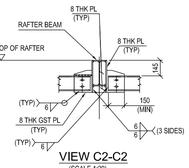


DETAIL-3
(SCALE 1:20)

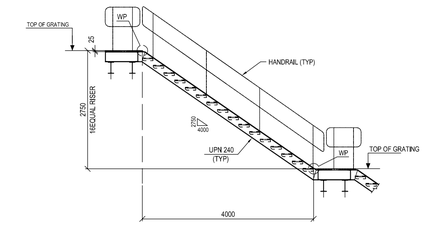
SECTION F-F
(SCALE 1:20)



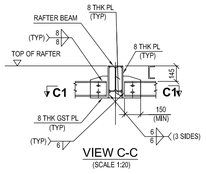
DETAIL-4
(SCALE 1:20)



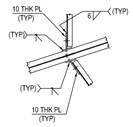
VIEW C2-C2
(SCALE 1:20)



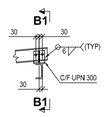
SECTION H-H
ELEVATION OF STAIR
(SCALE 1:50)



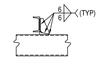
VIEW C-C
(SCALE 1:20)



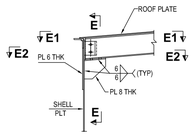
SECTION C1-C1
(SCALE 1:20)



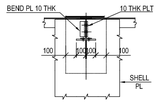
VIEW B-B
(SCALE 1:20)



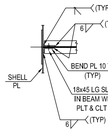
SECTION B1-B1
(SCALE 1:20)



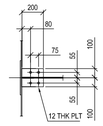
DETAIL-1
(SCALE 1:20)



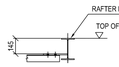
SECTION E-E
(SCALE 1:20)



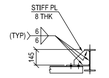
SECTION E1-E1
(SCALE 1:20)



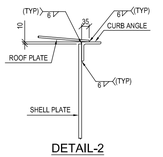
SECTION E2-E2
(SCALE 1:20)



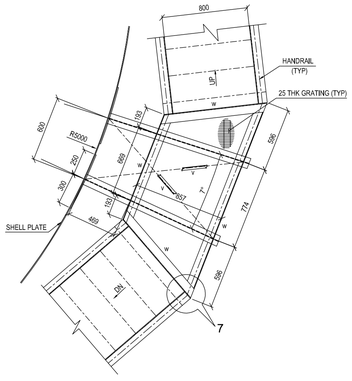
SECTION C3-C3
(SCALE 1:20)



SECTION C4-C4
(SCALE 1:20)



DETAIL-2
(TYP CONNECTION OF SHELL PLT ROOF PLT & CURB ANGLE)
(SCALE 1:10)


800
HANDRAIL (TYP)
25 THK GRATING (TYP)
R5500
250
600
250
300
150
469
SHELL PLATE
774
586
586
w
w
w